Slide 1 — Introduction

Good day everyone.

We are the University of South Africa team, and our project is called **CivicPulse Gauteng**.

Our project looks at voter participation at ward level, focusing specifically on Johannesburg and Tshwane, ahead of the 4 November 2026 Local Government Elections.

The main idea behind CivicPulse is to take electoral and community-level data, turn it into meaningful insights, and then make those insights usable for people who need to understand where and why participation differs.

Today, we will take you through the question we asked, the data we used, how we prepared it, what we found, the machine-learning model we developed, and finally how we turned the results into a usable application.

Slide 2 — The question

The question behind our project is:

To what extent do multi-dimensional deprivation and ward-level party dominance explain differences in voter turnout, especially among young voters, across Johannesburg and Tshwane?

We deliberately made the question more specific than simply asking why people do not vote.

We wanted to test two particular ideas.

1. The first is **deprivation friction**.

    The idea here is that wards experiencing poorer living conditions, such as poor water access, informal housing and irregular refuse collection, may experience lower voter participation.

2. The second is **incumbency demobilisation**.

    Here, we ask whether voters in wards where one party has held power for a long time may become less likely to participate, rather than switching to another party.

Importantly, we allowed the data to disagree with our assumptions. So a hypothesis not being supported is still considered a useful finding.

Slide 3 — Table of contents

To answer this question, we followed a complete data pipeline.

- We started with the introduction and our research question.
- We then moved into our data sources, cleaning and feature engineering.
- After that, we performed exploratory and statistical analysis before building and evaluating our models.
- Finally, we interpreted the results, discussed the limitations, and showed how the analysis can be deployed through CivicPulse.

Slide 4 — Objectives

We had four main objectives.

1. First, **map the metros**.
   - We wanted to create a ward-level picture of voter turnout across Johannesburg and Tshwane, so that organisations working on voter participation can see where differences occur.
2. Second, **test the drivers**.
   - We wanted to investigate whether service-delivery conditions and long-term party control were associated with lower participation.
3. Third, **earn trust**.
   - Instead of simply blaming young people for low participation, we wanted outreach to be grounded in evidence from individual wards.
4. And finally, **respect dignity**.
   - Voting is a civic right, and poor service delivery should not automatically mean that communities are written off.

Our analysis therefore focuses on understanding patterns rather than making assumptions about individual voters.

Slide 5 — Audiences

CivicPulse was designed with four audiences in mind:

- The first is the **IEC and civil society organisations**, who could use the information to identify areas where voter registration or turnout initiatives may need attention.
- The second is **political parties**, where the analysis can help investigate whether strongholds may be associated with different participation patterns.
- Third are **journalists**, who can use the data to go beyond the common explanation of simply calling low participation "youth apathy."
- And finally, **municipalities**, because we wanted to investigate whether conditions such as service delivery are reflected in patterns of civic participation.

With that, I will hand over to [NAME], who will explain the data and how we prepared it.

Slide 6 — Data cleaning and processing

Before we could analyse anything or build a model, we first had to make sure that the data itself was reliable.

We worked with the IEC election data and supplementary Census information.

One of the first problems we discovered was that the Census files we received were at **municipality level rather than ward level**.

This was important because our entire analysis is based on wards.

We also found a duplicate Johannesburg age file. We identified this by checking the file hashes rather than simply assuming that differently named files were different datasets.

The IEC data contained **195,424 ballot rows**.

After processing the data and removing presidential ballots, because those represent separate votes by the same people, we retained **45,086 rows** relevant to our analysis.

We also identified five voting districts where the number of votes exceeded the registered voting roll. Those values would produce impossible turnout figures, so we excluded them and documented the decision.

The important point is that we did not hide these problems. We identified them before modelling and recorded how they were handled.

Slide 7 — Feature engineering

After cleaning the data, we had to transform the raw information into variables that could actually be used for analysis.

For turnout, we calculated:

valid votes plus spoilt votes, divided by registered voters.

We also calculated the **effective number of parties**, or ENP.

This gives us a way of measuring political fragmentation. A lower value generally indicates that support is concentrated among fewer parties.

For representation, we created features such as the winning party, its vote share, the victory margin and individual party shares for the ANC, DA, EFF and ActionSA.

For the deprivation hypothesis, we used indicators relating to water, dwelling type and refuse collection to construct a standardised index.

We also incorporated changes between 2016 and 2021, including turnout change and incumbent-share change.

For youth, we used the Census cohort aged 14 to 25 in 2022 and aged that cohort forward to represent the potential 2026 electorate.

Finally, we created a geographic proxy using tent voting stations as an indicator of informal-settlement areas.

So every feature was created for a reason: it was connected to one of the questions we wanted to test.

Slide 8 — Exploratory analysis

Once the data was prepared, we moved into exploratory analysis.

One of the clearest findings was that **turnout appeared to split more strongly by party than by metro**.

Across the 242 wards, the mean ward turnout was **42.8%**, with turnout ranging from **21.1% to 65.7%**.

When we compared wards according to their winning party, the median turnout was **39.5% in ANC-won wards compared with 51.7% in DA-won wards**.

In comparison, the overall turnout figures for Johannesburg and Tshwane were **42.0% and 44.6%** respectively.

This suggested that the difference between the two metros was not the main story.

We also saw a relationship between party vote shares and turnout.

EFF vote share had a correlation of approximately **negative 0.78** with turnout, while DA vote share had a positive correlation of approximately **0.61**.

These findings helped us decide which political variables needed to be controlled for in our later modelling.

Slide 9 — Correlations

We then looked more closely at which variables moved together with turnout.

We used **Spearman correlations**, which are useful when the data does not necessarily meet the assumptions required for standard Pearson correlation.

Turnout and ANC vote share had a correlation of approximately **negative 0.63**, while turnout and DA vote share had a correlation of approximately **positive 0.61**.

We also found a very strong negative relationship between winning margin and ENP, at approximately **negative 0.92**.

This makes sense because a large winning margin generally means that political support is more concentrated.

Importantly, the relationship between the tent-station proxy and turnout was weak.

So, at this stage, we could not treat the tent-station indicator as strong evidence for our deprivation hypothesis.

Overall, **8 of the 9 feature pairs tested were statistically significant after false-discovery correction**.

Slide 10 — Statistical testing

We then moved from simply looking at patterns to formally testing them.

Because turnout was not normally distributed, we used the **Kruskal-Wallis test** to compare turnout across different winning-party groups.

The result was highly significant, with **H equal to 107.5 and a p-value of approximately 4.6 times 10 to the minus 24**.

This reinforced what we saw visually: party group was an important division in the turnout data.

We also used Spearman correlations with false-discovery-rate correction.

For the interaction model, we found that the relationship between dominance and turnout differed depending on the winning party.

For DA wards, the effect was **positive 0.41 standard deviations**, with p less than 0.001.

For ANC wards, the effect was much smaller, at **0.05 standard deviations**, and was not statistically significant.

Finally, our within-ward analysis of tent voting districts showed a difference of about **negative 1 percentage point**, but the confidence interval crossed zero.

So again, we could not treat that result as statistically significant.

And this is important: these are **associations within the data**, not proof that one factor causes another.

I will now hand over to [NAME] to explain our machine-learning models.

Slide 11 — Model selection

For the modelling stage, we treated turnout as a continuous value, so our primary task was **regression** rather than classification.

We compared several models.

Our baseline model had an MAE of **8.57 percentage points**.

OLS had an MAE of **6.33**, Ridge also had **6.33**, Random Forest had **6.28**, Gradient Boosting had **6.55**, and XGBoost had **6.50**.

At first glance, Random Forest performed slightly better.

However, we had already decided to use the **one-standard-error rule** when selecting the model.

Random Forest's advantage was small enough that Ridge fell within its one-standard-error range.

So we selected **Ridge regression**, because it gives us a simpler model while still providing comparable predictive performance.

This decision was made according to a rule set before looking at which model would be preferred, rather than simply choosing whichever number was lowest.

Slide 12 — What the model uses

We then looked at what the model actually relied on.

The strongest variable was **ANC vote share**.

When this variable was shuffled, the model's MAE increased by approximately **5.9 percentage points**, which indicates that the model relied heavily on it.

The next important variables were DA share, ENP and EFF share.

We also compared predicted turnout with actual turnout.

The held-out wards generally clustered around the diagonal line, meaning that the predictions followed the actual values reasonably well, although there was more variation at the extremes.

The model's average error was approximately **6.3 percentage points**.

That is important because the difference between the lowest and highest ward turnout was about **44 percentage points**.

So although the model is not perfect, it captures a meaningful amount of the variation in turnout.

Slide 13 — 2016 merge and forecasting

One of the important additions to the project was the use of the 2016 results.

This allowed us to move beyond a single-election snapshot.

When we used the 2016 turnout as a lagged feature to forecast 2021 turnout, the held-out MAE was **2.44 percentage points**, compared with **6.32 percentage points** for the cross-sectional model.

The R-squared also increased to **0.876**, compared with **0.365** in the cross-sectional model.

This tells us that historical turnout is a very strong signal for future turnout.

We then used this type of historical information to create a 2026 forecasting component.

For example, Ward 79900081 in Tshwane had 2021 turnout of **21.2%**, while the model's 2026 forecast was **11.9%**, with a 90% interval from **7.2% to 16.4%**.

We also created a mobilisation gap, which compares actual turnout with what the model expects for a comparable place.

One example had a gap of **negative 44.9 percentage points**.

The purpose is not to claim that the model knows exactly what will happen in 2026. Instead, it helps identify places where participation is substantially different from what the model expects.

Slide 14 — Interpretation

We then brought the results back to our original hypotheses.

For **H1, deprivation friction**, our current result is **inconclusive**.

The tent-station proxy showed approximately a one-percentage-point lower turnout within wards, but the result was not statistically significant.

The bigger problem is that our Census data is currently available at district or municipality level rather than the ward level needed for the full test.

So we do not claim that deprivation has been disproven.

For **H2, incumbency demobilisation**, the hypothesis was **not supported**.

In the cross-sectional analysis, DA strongholds showed higher turnout rather than the lower turnout expected under the hypothesis.

When we looked over time, turnout fell by **14.6 percentage points in retained wards** compared with **15.5 percentage points in wards that flipped**, and the difference was not statistically significant.

So the evidence did not show that simply holding a ward for longer was the reason for lower turnout.

Finally, our youth registration target is **not yet testable at ward level** because we still require the 2026 IEC registration data by age for individual wards.

Slide 15 — Evidence

This slide brings together the evidence supporting the conclusions we have just discussed.

The important point is that CivicPulse does not force every hypothesis into a positive result.

For deprivation, we report an inconclusive result because the evidence currently available is not sufficient for a ward-level conclusion.

For incumbency demobilisation, we report that the hypothesis was not supported by the tests we conducted.

And for youth participation, we clearly identify what data is still required before the question can be tested properly.

This distinction between **what the data shows and what the data cannot yet show** is an important part of our approach.

Slide 16 — Limits and honesty

There are several limitations that we wanted to make explicit.

1. First, **ward-level youth turnout is not published**, so we cannot directly model youth turnout.

- Instead, we separated the problem into overall turnout and youth registration.

2. Second, our Census export was at municipality level, which means that the deprivation hypothesis cannot yet be fully tested at ward level.

3. Third, ward boundaries changed between 2016 and 2021.

- That means that the same ward ID does not necessarily represent exactly the same geographical area.
- To address this, we used a voting-district crosswalk rather than simply matching the old and new ward IDs.
- We also have a relatively small sample of **242 wards**, which creates an overfitting risk.
- We addressed that through regularisation, grouped validation and additional voting-district-level checks.

4. Finally, there is an important ecological limitation.

- A pattern observed across wards does **not** automatically tell us how an individual voter behaves.

So our conclusions are about patterns in places, not assumptions about individual people.

Slide 17 — Deployment

The final part of the project was taking the analysis beyond a notebook.

CivicPulse is designed as a usable platform.

The pipeline exports the model and the data required by the application, including the trained model, model metadata, the master ward table and the 2026 forecast.

The application has three main views.

1. The first is the **Ward Explorer**, aimed at journalists and voters.
   - It allows users to browse information such as turnout, the winning party, margin, ENP and deprivation indicators.
2. The second is the **Priority List**, aimed at the IEC and civil society.
   - Instead of simply ranking wards by low turnout, it combines predicted turnout with youth-registration information to identify areas that may need further attention.
3. The third is the **Policy Simulator**, aimed at policymakers.
   - It allows users to change inputs and observe how the model's predicted turnout changes.

But we label these outputs as **associations rather than causes**, because the model does not prove that changing one factor will actually cause turnout to change by the predicted amount.

Slide 18 — Bonus techniques

We also included several techniques that go beyond a basic analysis.

The first is the **tent-station proxy**.

- We used station names in the IEC data to identify voting districts associated with informal settlements, giving us a sub-ward geographic signal without requiring another external dataset.

Second, we created a **2016-to-2021 crosswalk**.

- Instead of assuming that an old ward ID represents exactly the same place after boundaries changed, we re-aggregated the historical results onto the newer boundaries.

Third, we separated **defection from disengagement**.

- Rather than only looking at whether turnout decreased, we also examined whether the incumbent party's vote share changed.

Finally, we used the **one-standard-error and VIF rules** to make our model selection more disciplined.

These decisions helped us make the analysis more reproducible and less dependent on choosing the result that simply looked best.

Slide 19 — Final takeaway

So, what does CivicPulse ultimately tell us?

Our analysis shows that voter participation differs substantially across wards, and the clearest division in the current data is associated with political geography and party support rather than simply the difference between Johannesburg and Tshwane.

Our model can explain and predict part of this variation, with an average error of around **6.3 percentage points** in the cross-sectional model.

Historical turnout is particularly informative, with the 2016-to-2021 forecasting exercise producing substantially lower prediction error.

At the same time, we did not find enough evidence to confirm our deprivation hypothesis at ward level, and our incumbency-demobilisation hypothesis was not supported.

Most importantly, we believe the value of CivicPulse is not simply in producing one number.

It provides a way to move from raw electoral data, to evidence, to a tool that people can actually explore.

Slide 20 — Thank you

That brings us to the end of our presentation.

CivicPulse Gauteng demonstrates how public electoral and demographic data can be cleaned, analysed, modelled and turned into an interactive tool for understanding voter participation.

Thank you for listening.

We are happy to answer any questions.